XAI3 report
Interpretability of Bike Rentals and Real Estate Valuation Models

Adrianna Gryc

Universidad Politécnica de Valencia

Abstract. This report presents an interpretability analysis of machine learning models using Partial Dependence Plots (PDP). Two distinct datasets were analyzed: bike-sharing demand and King County housing sales. By applying model-agnostic methods, we isolated the marginal effects of key features, such as weather conditions and structural property attributes, to understand the decision-making process of underlying Random Forest models.

Keywords: XAI · Partial Dependence Plot · Random Forest · Model Interpretability.

1 Analyzing environmental drivers in Bike Sharing

The application of the PDP method allowed us to isolate the marginal effect of individual features on the predictions of the Random Forest model while keeping other variables constant.

The plot reveals a clear, step-like upward trend over time. The predicted number of rentals increases annually, with a sharp jump observed around day 400 (early 2012). This suggests that the bike-sharing system gained significant popularity during its second year of operation. The model learned that system maturity is a strong predictor of higher demand, independent of weather conditions.

We observe that the influence of temperature is non-linear. At low temperatures (below 5°C), the predicted number of rentals is low and stable (approx. 2,800). This is followed by a rapid surge in demand, peaking between 20°C and 25°C (over 5,000 rentals). Beyond 27°C, a decline is visible. This indicates an optimal comfort zone for cycling. Extreme cold discourages riding, but excessive heat also leads to a decrease in interest.

Moreover, the predicted rental count remains high when humidity is between 0% and approx. 60%. After crossing the 65% threshold, demand drops sharply, reaching its minimum near 100%. We can conclude that high humidity is a strong deterrent. As long as humidity remains moderate, it has little impact on user decisions, but extreme values significantly lower the model's prediction.

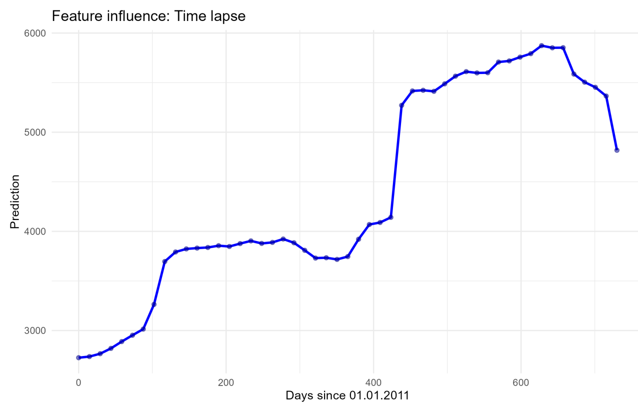

Fig. 1. Partial Dependence Plot for days since 2011.

Other important thing is that the plot shows a nearly linear decrease in predicted rentals as wind speed increases. Peak demand occurs during light winds (below 10 km/h), while speeds exceeding 25 km/h push the rental count below 3,700. Therefore strong wind is a negative factor that increases physical effort and lowers comfort. The model correctly identified that more difficult atmospheric conditions lead to fewer people choosing a bike.

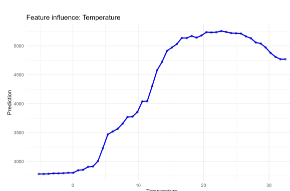

Fig. 2. PDP for Temperature.

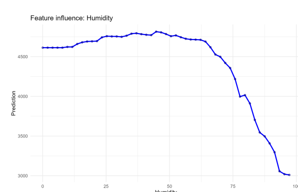

Fig. 3. PDP for Humidity.

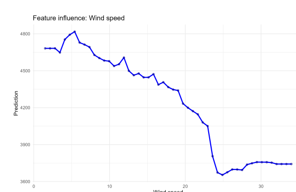

Fig. 4. PDP for Wind Speed.

2 Interactions between temperature and humidity

The 2D plot allows for an understanding of the complex relationship between two features that cannot be fully captured by 1D plots. The Random Forest model identified clear zones of influence for weather conditions.

The highest predicted values occur within a broad temperature window from 15°C to 30°C and humidity levels between 40% and 70%. In this range, the impact of both variables is stable, suggesting that demand is resilient to minor weather fluctuations within these boundaries.

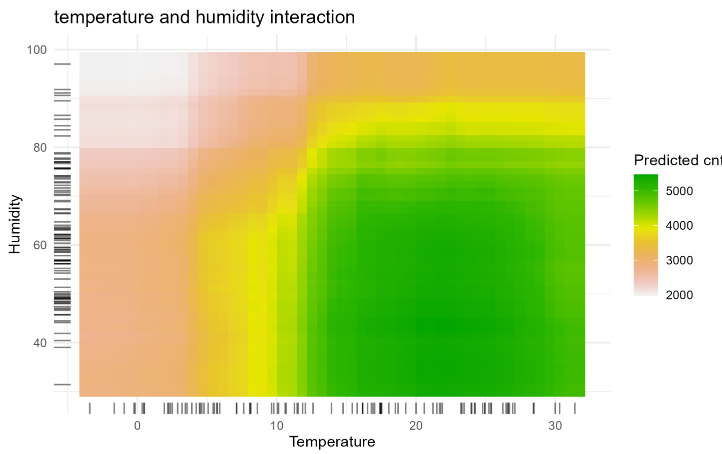

Fig. 5. 2D Partial Dependence Plot of Temperature and Humidity.

Critical impact of high humidity

At humidity levels exceeding 80% the predicted number of rentals drops sharply, regardless of the temperature. Even on a day with an ideal temperature of 25°C, very high humidity drastically reduces the attractiveness of cycling as a mode of transport.

Low temperature and high humidity synergy

The lowest forecasts are visible in the upper-left corner of the plot, where the temperature drops below 5°C and humidity is very high. There is a negative synergy here, as the combination of cold and moisture is the most discouraging for users. Interestingly, at very low temperatures a decrease in humidity does not cause as rapid a surge in demand as it does at higher temperatures. This suggest that cold is a more dominant limiting factor than humidity.

Data distribution analysis

The rug plots on the axes show that most observations in the dataset are concentrated between 10°C and 25°C and humidity above 40%. The model is most reliable in the green and yellow zones where it has the most training examples. Interpretations in areas with low data density (e.g., humidity below 30%) should be treated with caution.

3 Structural Ddterminants in Real Estate pricing

The PDP method enabled an investigation into how the predicted house price changes under the influence of selected features while averaging out the impact of all other input variables.

Living space is the most critical predictor in the model. The plot shows a nearly linear upward trend. The predicted price rises sharply from approx. $400,000 for small homes to over $1,250,000 for residences with 8,000 sqft. The market consistently values square footage. The steps on the plot may suggest psychological price points or specific market segments (e.g., luxury homes above 4,000 sqft).

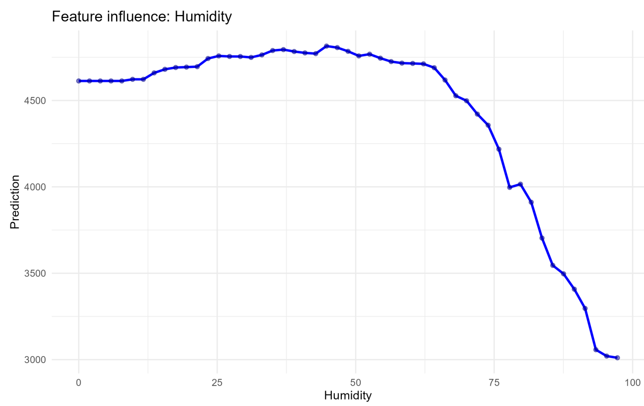

Fig. 6. Impact of Living Area (sqft) on predicted price.

When it comes to the number of bathrooms there is a strong positive correlation. The most significant price increase occurs in the range of 3 to 5 bathrooms. Beyond 5 bathrooms, the curve flattens. The model identified that a large number of bathrooms is a trait of high-standard homes, but after reaching a certain threshold, each additional bathroom adds less marginal value to the property.

In the number of floors the relationship is not simple. The price increases for 1.5-story homes compared to single-story ones, but there is a slight dip for exactly 2-story homes. The highest valuations are generated for 2.5 and 3-story houses. Such homes may be perceived as more unique or modern (e.g., lofts), which translates to a higher unit price despite lower overall popularity.

The plot for number of bedrooms provides the most surprising results. The highest predicted price is for 1-bedroom homes, which then drops for 2 and 3-bedroom properties. It likely stems from the fact that 1-bedroom homes in this dataset are often luxury apartments in prestigious, high-density urban locations (e.g., city centers). In contrast, homes with a very high number of bedrooms might be older multi-family buildings in less desirable locations, which lowers their average predicted value when other features are neutralized.

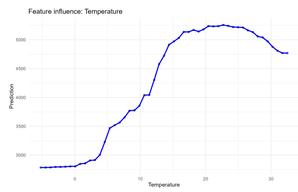

Fig. 7. PDP for Bathrooms.

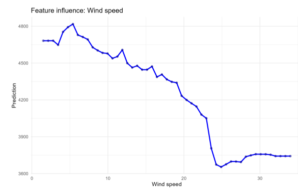

Fig. 8. PDP for Floors.

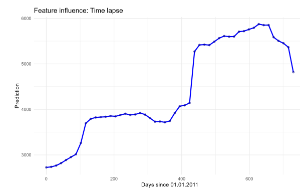

Fig. 9. PDP for Bedrooms.

4 Final summary

The analysis confirms that the Random Forest models have successfully captured logical and intuitive relationships across both datasets. In the bike-sharing study, we identified clear optimal weather conditions and significant deterrents like extreme humidity. In the real estate study, while living space is the dominant factor, the model also reveals complex non-linearities in features like bedroom count and floor levels. These insights demonstrate that XAI methods like PDP are essential for validating black-box models and extracting actionable knowledge from data.